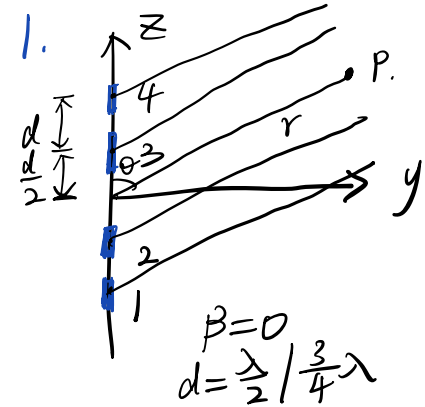




a) The radiated field of a dipole at origin $O =$

$$\vec{E}_0(r, \theta) = \vec{a}_\theta \frac{j\eta K I_0 L}{4\pi r} e^{-jkr} \sin\theta \quad (\vec{I} = \vec{a}_z I)$$

The distance of these 4 dipoles =

For amplitude = $r_1 = r_2 = r_3 = r_4$

For phase = $r_1 = r + \frac{3}{2}d \cos\theta$

$r_2 = r + \frac{1}{2}d \cos\theta$

$r_3 = r - \frac{1}{2}d \cos\theta$

$r_4 = r - \frac{3}{2}d \cos\theta$

Total Electric field =

$$\vec{E}(r, \theta) = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4$$

$$= \vec{a}_\theta \frac{j\eta K I_0 L}{4\pi r} e^{-jkr(r + \frac{3}{2}d \cos\theta)} \sin\theta + \vec{a}_\theta \frac{j\eta K I_0 L}{4\pi r} e^{-jkr(r + \frac{1}{2}d \cos\theta)} \sin\theta +$$

$$+ \vec{a}_\theta \frac{j\eta K I_0 L}{4\pi r} e^{-jkr(r - \frac{1}{2}d \cos\theta)} \sin\theta + \vec{a}_\theta \frac{j\eta K I_0 L}{4\pi r} e^{-jkr(r - \frac{3}{2}d \cos\theta)} \sin\theta$$

$$= \vec{a}_\theta \frac{j\eta K I_0 L}{4\pi r} e^{-jkr} \sin\theta \left[e^{j\frac{1}{2}kd \cos\theta} + e^{j\frac{1}{2}kd \cos\theta} + e^{-j\frac{1}{2}kd \cos\theta} + e^{-j\frac{3}{2}kd \cos\theta} \right]$$

$$= \vec{a}_\theta \frac{j\eta K I_0 L}{4\pi r} e^{-jkr} \sin\theta \cdot 2 \left[\cos\left(\frac{1}{2}kd \cos\theta\right) + \cos\left(\frac{3}{2}kd \cos\theta\right) \right]$$

$$= \vec{a}_\theta \frac{j\eta K I_0 L}{4\pi r} e^{-jkr} \sin\theta \cdot 4 \cos(kd \cos\theta) \cdot \cos\left(\frac{1}{2}kd \cos\theta\right)$$

$$= \vec{E}_0(r, \theta) \cdot AF(\theta)$$

↓ because
 $\cos 3x + \cos x = 2 \cos 2x \cdot \cos x$

The array factor $AF(\theta)$ is same as the result of N-element formula =

$$|AF(\theta)| = \left| \frac{\sin(N\psi/2)}{\sin(\psi/2)} \right| \quad N=4 \quad \left| \frac{\sin(2\psi)}{\sin(\psi/2)} \right| = \left| \frac{2 \sin(\psi/2) \cos(\psi/2) \cos \psi}{\sin(\psi/2)} \right|$$

$$= 2 |\cos \psi \cdot \cos(\psi/2)| \quad \psi = kd \cos\theta$$

$$= 2 |\cos(kd \cos\theta) \cdot \cos\left(\frac{1}{2}kd \cos\theta\right)|$$

when $d = \frac{\lambda}{2}$, $kd = \pi$ =

$$\vec{E}(r, \theta) = \vec{a}_\theta \frac{j\eta K I_0 L}{4\pi r} e^{-jkr} \sin\theta \cdot 4 \cos(\pi \cos\theta) \cdot \cos\left(\frac{\pi}{2} \cos\theta\right)$$

when $d = \frac{3}{4}\lambda$, $kd = \frac{3}{2}\pi =$

$$\vec{E}(r, \theta) = \vec{A}_\theta \frac{j\eta k I_0 \Delta L}{4\pi r} e^{-jkr} \sin\theta \cdot 4 \cos\left(\frac{3}{2}\pi \cos\theta\right) \cdot \cos\left(\frac{3}{4}\pi \cos\theta\right)$$

b) c) d)

When $d = \frac{\lambda}{2}$, $E(\theta) = F(\theta) \cdot AF(\theta) = \sin\theta \cdot 4 \left| \frac{\sin(4\psi)}{\sin(\psi/2)} \right|$, $\psi = \pi \cos\theta$

For $F(\theta)$, $\theta_{\max} = \frac{\pi}{2}$

$\theta_{\text{null}} = 0, \pi$

For $AF(\theta)$, $\theta_{\max} = \cos^{-1} \left[\frac{1}{kd} (-\beta \pm 2n\pi) \right]$ $n=0$ for main beam

$$= \cos^{-1} \left[\frac{1}{\pi} (\pm 2n\pi) \right] = \cos^{-1}(0)$$

$$= \frac{\pi}{2}$$

$$\theta_{\text{null}} = \cos^{-1} \left[\frac{1}{kd} (-\beta \pm \frac{2n\pi}{N}) \right] \quad n=1, 2, \dots$$

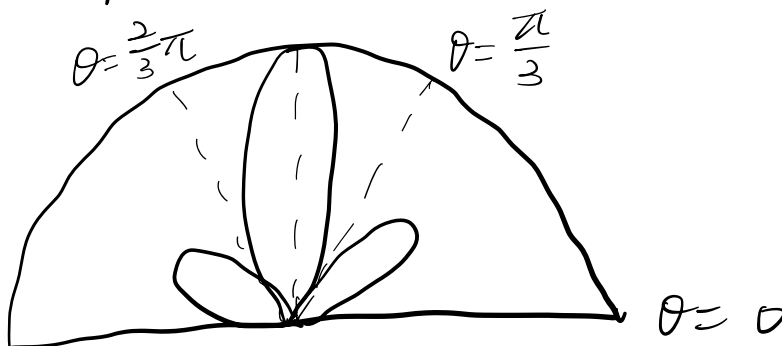
$$= \cos^{-1} \left[\frac{1}{\pi} (\pm \frac{n\pi}{2}) \right] = \cos^{-1}(\pm \frac{1}{2}), \cos^{-1}(\pm 1)$$

$$= 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi$$

So $d = \frac{\lambda}{2}$, main beam direction is $\theta = \frac{\pi}{2}$

null directions are $\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi$

radiation pattern =



$\theta = \pi$

$\theta = 0$

when $d = \frac{3}{4}\lambda$, $E(\theta) = F(\theta) \cdot AF(\theta) = \sin \theta \cdot 4 \left| \frac{\sin(4\psi)}{\sin(\psi/2)} \right|$, $\psi = \frac{3}{2}\pi \cos \theta$

For $F(\theta)$, $\theta_{\max} = \frac{\pi}{2}$

$\theta_{\text{null}} = 0, \pi$

For $AF(\theta)$, $\theta_{\max} = \cos^{-1} \left[\frac{2}{3\pi} (\pm 2n\pi) \right]$ $n=0$

$= \frac{\pi}{2}$

$\theta_{\text{null}} = \cos^{-1} \left[\frac{2}{3\pi} (\pm \frac{n\pi}{2}) \right]$ $n=1, 2, \dots$

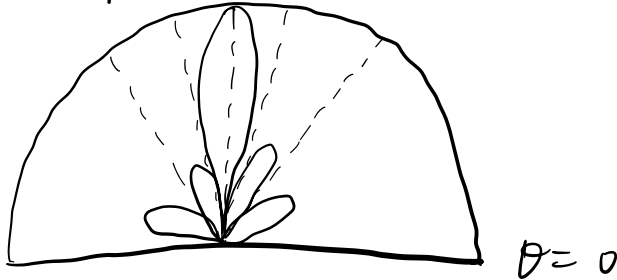
$= \cos^{-1} (\pm \frac{1}{3}), \cos^{-1} (\pm \frac{2}{3}), \cos^{-1} (\pm 1)$

$= 70.5^\circ, 109.5^\circ, 48.2^\circ, 131.8^\circ, 0, \pi$

So $d = \frac{3}{4}\lambda$, main beam direction $= \theta = \frac{\pi}{2}$

null direction $= \theta = 0, 48.2^\circ, 70.5^\circ, 109.5^\circ, 131.8^\circ, \pi$

radiation pattern =



$\theta = 0$

e) when $d = \frac{\lambda}{2}$

$\theta_{\max} = 45^\circ = \cos^{-1} \left[\frac{1}{\pi} (-\beta) \right]$

$\beta = -\frac{\sqrt{2}}{2}\pi$

when $d = \frac{3}{4}\lambda$

$\theta_{\max} = 45^\circ = \cos^{-1} \left[\frac{2}{3\pi} (-\beta) \right]$

$\beta = -\frac{3\sqrt{2}}{4}\pi$

f) when $d = \frac{\lambda}{2}$, no grating lobes for any β .

when $d = \frac{3}{4}\lambda$, in order to avoid grating lobes,

$0 < \beta < \pi, \frac{3\pi}{2} + \beta < 2\pi$
 $\pi < \beta < 2\pi, \frac{3\pi}{2} - \beta < 0$

$0 < \beta < \frac{\pi}{2}$ or $\frac{3\pi}{2} < \beta < 2\pi$

2. use two vertical elements as a array first,

$$|AF(\psi)| = 2 |\cos(\psi/2)|, \quad d = \frac{\lambda}{4}, \quad \beta = \frac{\pi}{2}, \quad \psi = \frac{\pi}{2} \cos \theta + \frac{\pi}{2}$$

$$|AF(\theta)| = 2 |\cos(\frac{\pi}{4} \cos \theta + \frac{\pi}{4})|$$

then, use 4 horizontal elements as a array =

$$|AF(\psi')| = 4 \left| \frac{\sin(2\psi')}{\sin(\psi'/2)} \right| = 4 \left| \frac{2 \sin(\psi') \cos(\psi')}{\sin(\psi'/2)} \right| = 16 |\cos(\psi') \cos(\psi'/2)|$$

$$d = \frac{\lambda}{2}, \quad \beta = 0, \quad \psi' = \pi \cos \phi$$

$$|AF(\phi)| = 16 |\cos(\pi \cos \phi) \cos(\frac{\pi}{2} \cos \phi)|$$

Total array factor =

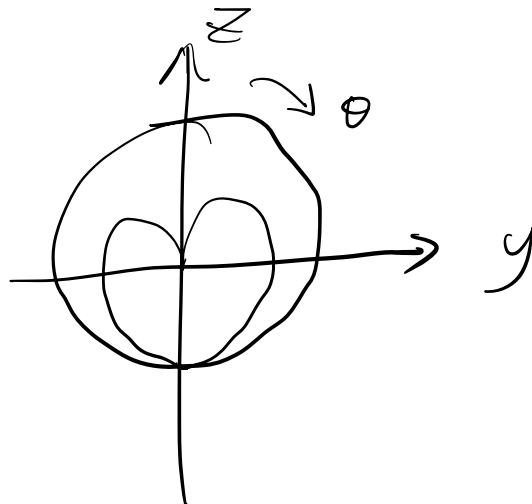
$$\begin{aligned} |AF(\theta, \phi)| &= |AF(\theta)| |AF(\phi)| \\ &= 32 \left| \cos(\frac{\pi}{4} \cos \theta + \frac{\pi}{4}) \cos(\pi \cos \phi) \cos(\frac{\pi}{2} \cos \phi) \right| \end{aligned}$$

For $x \rightarrow z$, $\phi = 0^\circ$, $|AF(\theta, 0)| = 0$, no pattern

$$\text{For } y \rightarrow z, \phi = 90^\circ, |AF(\theta, 90^\circ)| = 32 \left| \cos(\frac{\pi}{4} \cos \theta + \frac{\pi}{4}) \right|$$

$$\theta_{\max} = \pi$$

$$\theta_{\text{null}} = 0$$



For xoy , $\theta = 90^\circ$, $|AF(90, \phi)| = 16\sqrt{2} |\cos(\pi \cos \phi) \cos(\frac{\pi}{2} \cos \phi)|$

$$|\cos(\pi \cos \phi_{max})| = 1, \phi_{max} = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$$

$$|\cos(\frac{\pi}{2} \cos \phi_{max})| = 1, \phi_{max} = \frac{\pi}{2}, \frac{3\pi}{2}$$

Total $\phi_{max} = \frac{\pi}{2}, \frac{3\pi}{2}$

$$|\cos(\pi \cos \phi_{null})| = 0, \phi_{null} = 60^\circ, 120^\circ, 240^\circ, 300^\circ$$

$$|\cos(\frac{\pi}{2} \cos \phi_{null})| = 0, \phi_{null} = 0, \pi, 2\pi$$

Total $\phi_{null} = 0, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ, 360^\circ$

